

(Chapter #9)

Counter :-

- It is sequential circuit.
- Counter is a digital device used to count number of pulses and it can also be used as frequency division.
- Counter count in two

ways:

- ★ 1 up count (0, 1, 2, ..., N)
- ★ 2 down count (N, N-1, ..., 1, 0)

→ Present count of the counter represents state of counter.

→ Counter contains set of flip-flops, A n-bit counter will have N

flip flops & 2^n states.

→ Each state frequency = $\frac{\text{total frequency}}{2^n}$

Counter

Asynchronous

Synchronous

+ , + Down counting
- , + up counting
(1)

Asynchronous Counters

→ A synchronous refers to states that does not have a fixed relationship with each other.

→ In Asynchronous counters flip flops will have common clock pulse, so their states does not change exactly same time.

→ 1st flip-flop has clock, output of each flip flop will acts as clock to the next flip-flop in the counter. T and JK flip-flops are used.

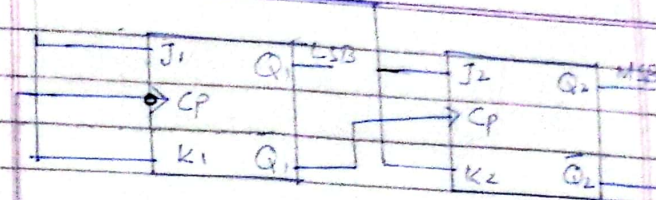
Example:-

Ripple counter

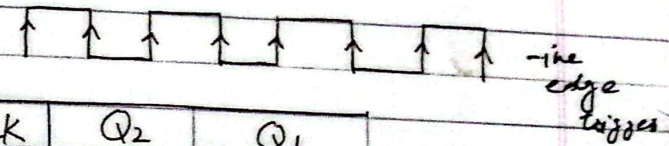
2-bit Ripple-up-Counter

(2)

high voltage (Logic)



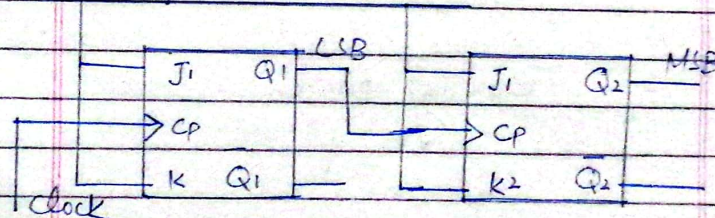
clock



CLK	Q ₂	Q ₁
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

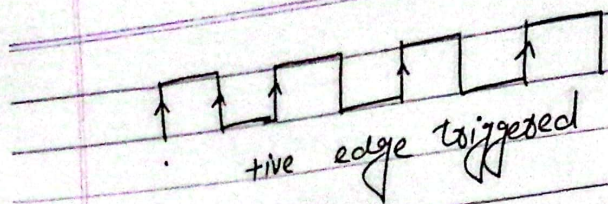
2-bit Ripple Down Counter

high voltage (Logic)



clock

(3)



CLK	Q ₂	Q ₁	
0	0	0	3
1	1	1	2
2	1	0	1
3	0	1	0
4	0	0	0

Applications

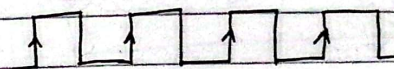
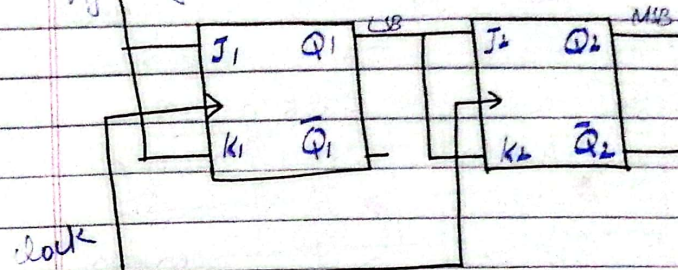
Counter Application

- Counting events
- System clock
- Times, delays
- Sequence control
- Frequency Division

(4)

Synchronous flip flop

High (logic) (up counter)



CLK	Q ₂	Q ₁	∴ Synchronous means at the same time.
0	0	0	
1	0	1	
2	1	0	
3	1	1	
4	0	0	

- All flip flops are synchronized to the same clock input.
- Means for every clock pulse, all the flip-flops will generate an output.
- JK-flip-flop, D flip-flop or T-flip-flop to make synchronous counters.

(5)

(10)

Differentiate

Synchronous counters

→ All flip flops are triggered with same clock simultaneously.

→ Synchronous is faster.

→ It does not produce any decoding errors.

→ Also called parallel counter.

→ Designing is very complex due to increasing the number of states.

→ will operate in any desired count sequence.

→ e.g., Ring counter, Johnson counter.

Asynchronous counters

→ Different flip flop are triggered with different clock, not simultaneously.

→ It is slower.

→ It produces decoding errors.

→ Also called serial counter.

→ Designing as well as implementation is very easy.

→ Asynchronous counter operate only in final count sequence (Up/Down).

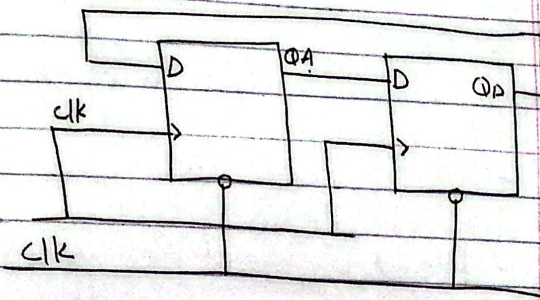
e.g., Ripple up counter, Ripple Down counter

(6)

number of FFs

Ring Counter :-

It works on the principle of feedback.



It is constructed using D-flip flop.

In D flip flop

$D=0$, o/p = 0 (reset)

$D=1$, o/p = 1 (set)

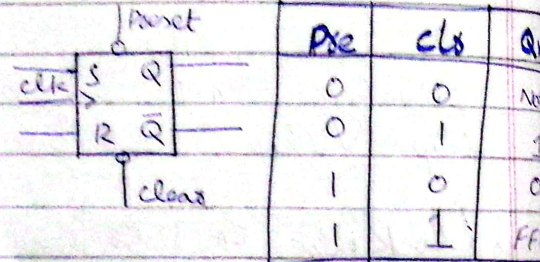
Initially, $QA=1$ and other flip flops $QB=0$

After Clock pulse	CA	CB
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

(7)

What are Preset and clear input of flip flops.

- `Pnext` and `clear` inputs are also called crosscutting or Asynchronous inputs.
- `Pnext` and `clear` gives predefined output whenever other inputs `clear`.
- `Pnext` and input does not need synchronization.
- `Pnext` is used to make output 1.
- `clear` is used to make output $Q_n = 0$.

$$P_{\text{reset}} = 0 \Rightarrow Q_n = 1$$
$$\text{clear} = 0 \Rightarrow Q_n = 0$$


There are active low inputs.

Shift register Counters

Definition:-

A shift register counter is basically a shift register with the serial output connected back to the serial input to produce special sequences.

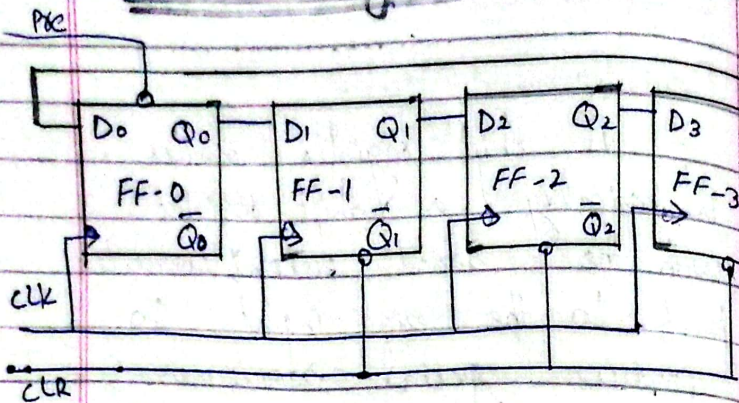
These devices are often classified as counters because they exhibit a specified sequence of states.

Types: Two most common types
of shift registers counters
The Johnson counter
The ring counter

The Ring Counter:-

- Ring counter is a typical application of shift register.
- The O/P of the last flip flop is connected to the I/P of first flip flop.
- (a)

Circuit Diagram :-



$\Rightarrow$ No of states = No of ffs or bits

Truth table:-

$\text{Clr} = 0 \rightarrow \text{if } 0/P = 0$
 $\text{Pre} = 0 \rightarrow \text{if } 0/P = 1$
 Active $\therefore$ No of states = 16

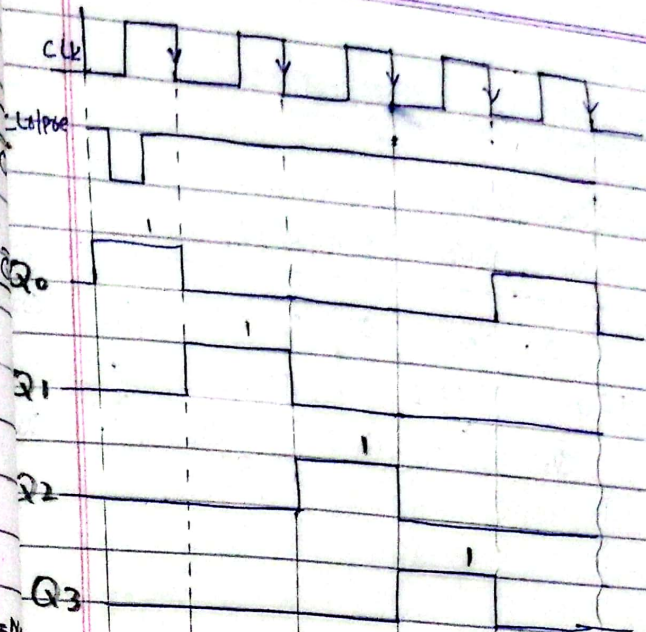
clk	Pre	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	x		1	0	0	0
1	↓		0	1	0	0
1	↓		0	0	1	0
1	↓		0	0	0	1
1	↓		1	0	0	0

Purpose of Clear/Pre:-

- Active low inputs, Asynchronous inputs, overriding inputs
- When we activate them only preset/Clr will work its no matter what is the input

number of FFs

wave form:-



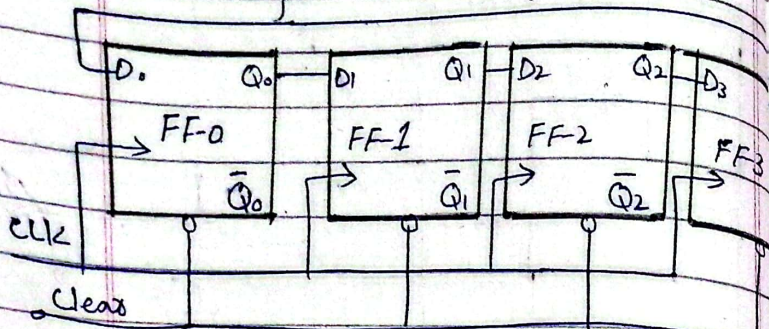
Johnson Counter

- Also called twisted ring counter / Tail ring counter
- In Johnson counter the complement of the ^{last} flip-flop is connected back to D input of the first flip flop.

(11)

Circuit Diagram:-

- No Preset is required
- No of states = $2 \times \text{no of fl}$

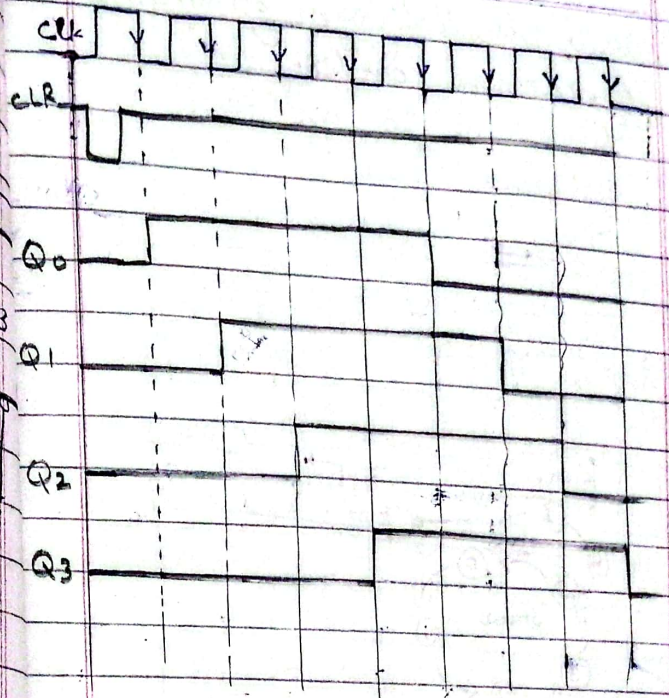


Truth table:-

CLK	clock	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	1
1	↓	0	0	0	0

(1a)

Wave form:-



Q# What is Modulus or Mod of Counter?

→ The no of states passed by the counter before going to its initial state is called modulus or mod of counter. $N \leq 2^n$

for example:

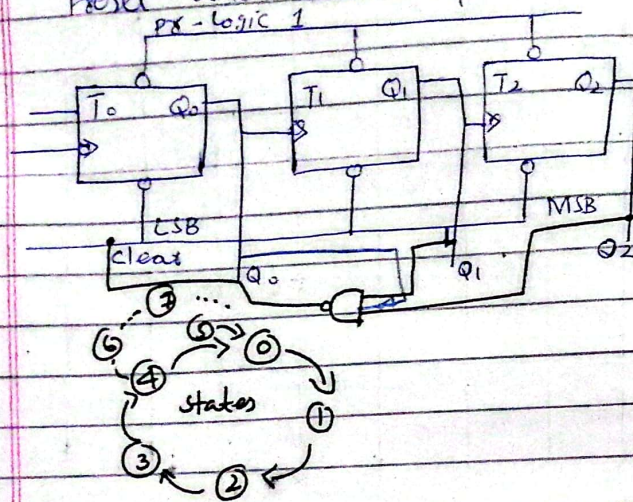
if we have used 4 flip-flops then no of states

$$N = 2^n = 2^4 = 16 \text{ states}$$

then we say this mod 16 counter

Draw circuit diagram of MOD-6 Asynchronous counter.

We have to use 3-bit Asynchronous counter and use Reset and clear input.



Describe decade ripple counter.

Decade (BCD) ripple counter

Mod-10 $\Rightarrow$ states = $10 = 2^n$

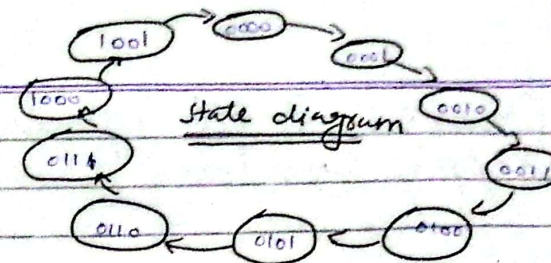
Max-count = $2^n - 1 = 10 - 1 = 9$

BCD 0 $\rightarrow$ 9

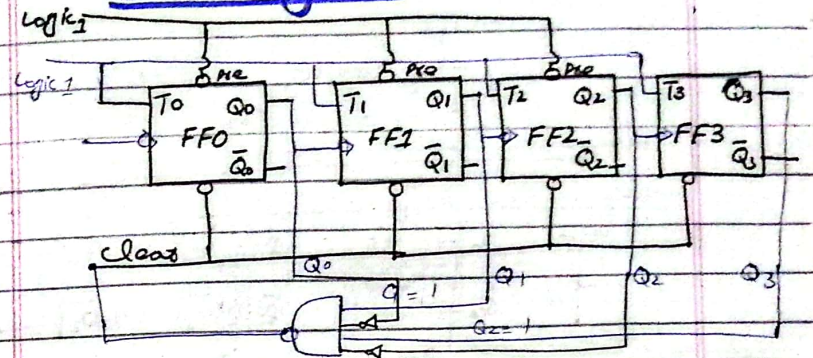
0000 $\rightarrow$ 1001

4-bits 4 flip-flops

$2^4 = 16$ states
max-count = $16 - 1 = 15$
(16) 0 to 15 $\Rightarrow$ 0 to 15



Block diagram:



Applications: State machines, clock division, clock circuits, clock generation, Sequencers

Truth table :-

Dec	D	Q ₃	Q ₂	Q ₁	Q ₀
0	0	0	0	0	0
1	1	0	0	0	1
2	2	0	0	1	0
3	3	0	0	1	1
4	4	0	1	0	0
5	5	0	1	0	1
6	6	0	1	1	0
7	7	0	1	1	1
8	8	1	0	0	0
9	9	1	0	0	1
10	10	1	0	1	0
11	11	1	0	1	1
12	12	1	1	0	0
13	13	1	1	0	1
14	14	1	1	1	0
15	15	1	1	1	1

$clk = \overline{Q_3} \cdot Q_1$

(15)

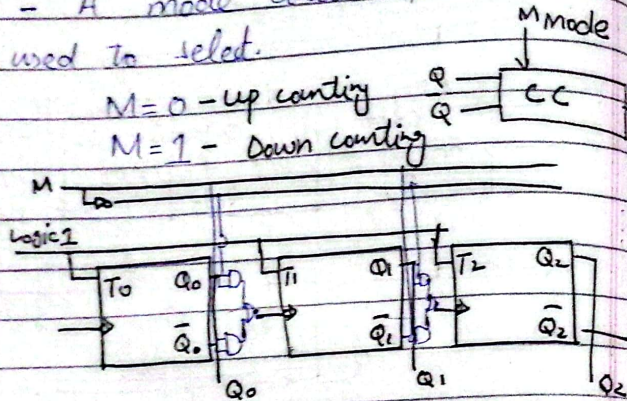
Describe 3-bit up-down Ripple Counter.

- Both up and down modes are combined.

- A mode control i/p (M) is used to select.

M=0 - up counting

M=1 - Down counting



M	Q	$\bar{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$Y = \bar{M}Q + M\bar{Q}$$

Date _____

Day MTWTFSS

3-bit Synchronous Design =

Step 1:- Decide the no and types of flip-flops
3-bit, FFs=3, JK

Step 2:- Design excitation table.

Qn	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

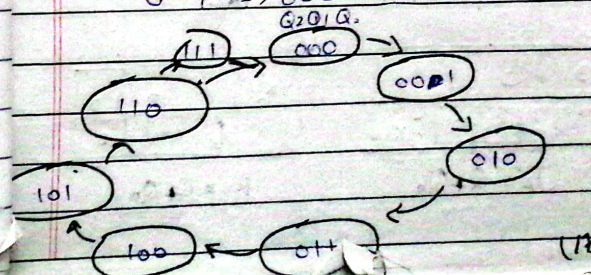
Step 3:-

State diagram and ckt Exc table

$2^3 = 8$ states

max count = $8-1 = 7$

$0-7 \Rightarrow 000 - 111$



NS	$Q_2 Q_1 Q_0$	$J_0 K_0$	$J_1 K_1$
000	001	1 X	0 X
001	010	X 1	1 X
010	011	1 X	X 0
011	100	X 1	X 1
100	101	1 X	0 X
101	110	X 1	1 X
110	111	1 X	X 0
111	000	X 1	X 1

Step 4:-

$Q_2 Q_1 Q_0$	J_0	J_1	J_2
000	1	1	1
001	1	1	1
010	1	1	1
011	1	1	1

$J_0 = 1$

$J_1 = Q_0$

$Q_2 Q_1 Q_0$	K_0	K_1	K_2
000	1	1	1
001	1	1	1
010	1	1	1
011	1	1	1

$K_0 = 1$

$K_1 = Q_0$

J_2

$Q_2 Q_1 Q_0$	J_2
000	0
001	0
010	0
011	0
100	0
101	0
110	0
111	0

$J_2 = Q_1 + Q_0$

K_2

$Q_2 Q_1 Q_0$	K_2
000	0
001	0
010	0
011	0
100	0
101	0
110	0
111	0

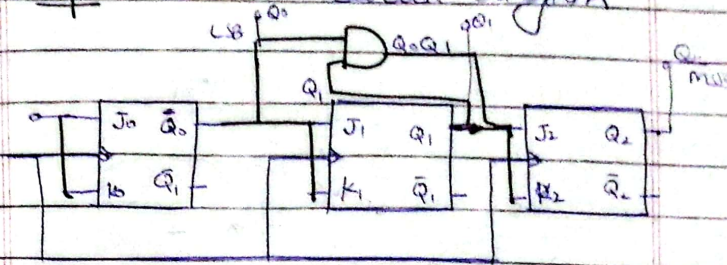
$K_2 = Q_1 + Q_0$

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_0 + Q_1$

Step 5:- Draw Circuit diagram



MOD-6 Down Synchronous Counter

Step 1:-

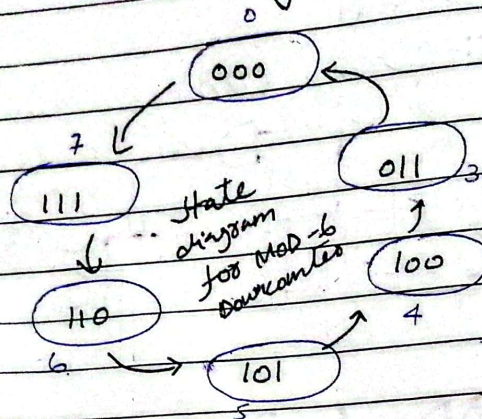
Mod-6 $\Rightarrow$ 3-bits = 3 FFs, JK FF

Step 2:-

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Date _____

Step 3: State diagram



Step 4:-

Excitation table for circuit

PS $Q_2 Q_1 Q_0$	NS $Q_2 Q_1 Q_0$	J ₂ K ₂		J ₁ K ₁		J ₀
000	111	1	X	1	X	1
111	110	X	0	X	0	X
110	101	X	0	X	1	1
101	100	X	0	0	X	X
100	011	X	1	1	X	1
011	000	0	X	X	1	X

J ₂				K ₂			
Q ₂ Q ₁	01	11	10	Q ₂ Q ₁	01	11	10
0	1	X	X	0	X	X	0
1	X	0	X	1	X	X	0

$J_2 = \bar{Q}_0$ $K_2 = \bar{Q}_1 \bar{Q}_0$

Date _____

J ₁				K ₁			
Q ₂ Q ₁	01	11	10	Q ₂ Q ₁	01	11	10
0	1	X	X	0	X	X	1
1	X	X	0	1	X	1	X

$J_1 = \bar{Q}_0$ $K_1 = \bar{Q}_2 + \bar{Q}_0$

J ₀				K ₀			
Q ₂ Q ₁	01	11	10	Q ₂ Q ₁	01	11	10
0	1	1	1	0	X	X	X
1	X	X	X	1	1	1	1

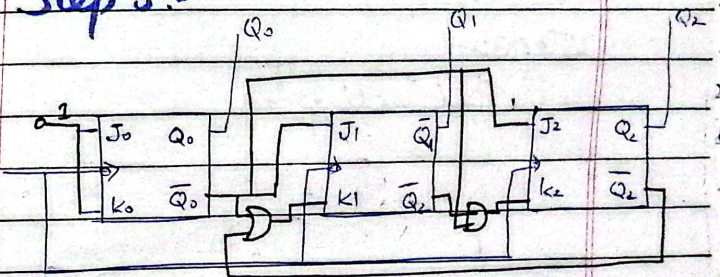
$J_0 = 1$ $K_0 = 1$

$$\therefore J_0 K_0 = 1$$

$$\therefore J_2 K_2 = \bar{Q}_1 \bar{Q}_0$$

$$J_1 = \bar{Q}_0, K_1 = \bar{Q}_2 + \bar{Q}_0$$

Step 5:-



Date _____

Day [M][T][W][T][F]

Problem: Design a Synchronous counter that has the following sequence 000, 010, 101, 110 and repeat

Step 1:-

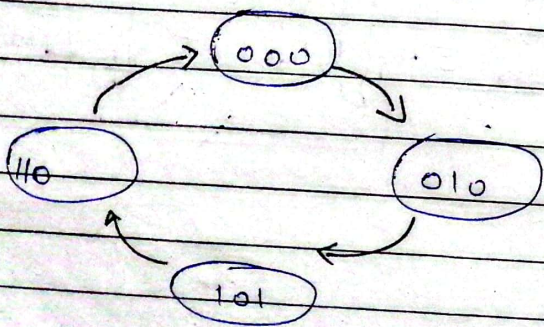
No of FFs = no of states
= 3 FFs, D-type FF

Step 2:-

Q_n	Q_{n+1}	D	Excitation table
0	0	0	
0	1	1	
1	0	0	
1	1	1	

Step 3:-

State diagram



(22)

Step 4:-

PS	NS	I/P
$Q_2 Q_1 Q_0$	$Q_2 Q_1 Q_0$	$D_2 D_1 D_0$
000	010	0 1 0
001	xxx	xxx
010	101	1 0 1
011	xxx	xxx
100	xxx	xxx
101	110	1 1 0
110	000	0 0 0
111	xxx	xxx

D_2

$Q_2 Q_1$	00	01	11	10
Q_0	0	1	0	x
1	x	x	x	1

$$D_2 = Q_0 + \bar{Q}_2 Q_1$$

D_1

$Q_2 Q_1$	00	01	11	10
Q_0	1	0	0	x
1	x	x	x	1

$$D_1 = \bar{Q}_1 + Q_0$$

$Q_2 Q_1$	00	01	11	10
Q_0	0	1	0	x
1	x	x	x	0

$$D_0 = \bar{Q}_2 Q_1$$

$$D_0 = \bar{Q}_2 Q_1$$

$$D_1 = \bar{Q}_1 + Q_0$$

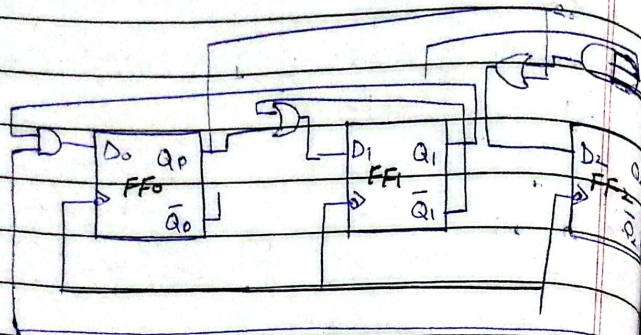
$$D_2 = Q_0 + \bar{Q}_2 Q_1$$

(23)

Date _____

Day M T W T F S S

Step 5:-



3-Bit Synchronous up Down Counter

Step 1:-

$n=3$ bits

$FF = T$

T flip flop

Step 2:-

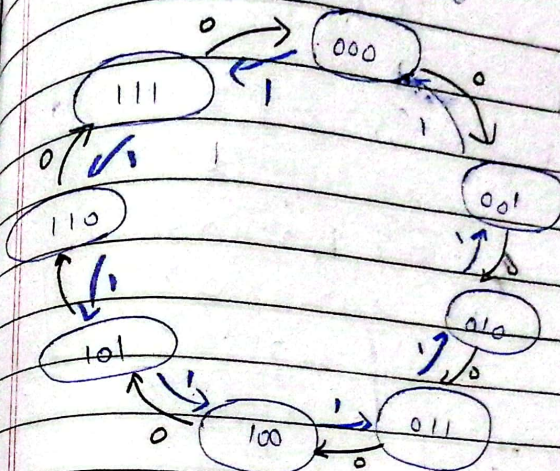
Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

(24)

Step 3:-

$M=0$ up count
 $M=1$ down count



Step 4

	PS $Q_2 Q_1 Q_0$	NS $Q_2 Q_1 Q_0$	$T_2 T_1 T_0$
0	0 0 0	0 0 1	0 0 1
0	0 0 1	0 1 0	0 1 1
0	0 1 0	0 1 1	0 0 1
0	0 1 1	1 0 0	1 1 1
0	1 0 0	1 0 1	0 0 1
0	1 0 1	1 1 0	0 1 1
0	1 1 0	1 1 1	0 0 1
0	1 1 1	0 0 0	1 1 1

(25)

Data

Day (m) (T) (w) (T) (P) (S)

1	0	0	0	1	1	1	1	1
1	0	0	1	1	1	0	1	1
1	0	1	0	1	0	1	0	1
1	0	1	1	1	0	0	1	1
1	1	0	0	0	1	1	0	1
1	1	0	1	1	1	1	1	1
1	1	1	0	0	0	0	0	1
1	1	1	1	0	0	1	0	1
1	0	0	1	0	0	0	0	1

$T_0 = 1$

T_1 MQ_2

Q_1, Q_0	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1

$T_1 = Q_0 \bar{M} + \bar{Q}_0 M$

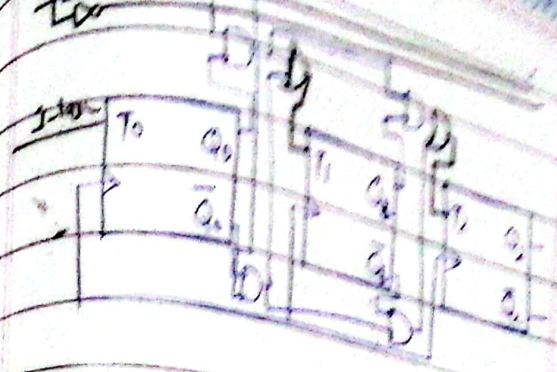
T_2 MQ_2

Q_1, Q_0	00	01	11	10
00	0	0	1	1
01	0	0	0	0
11	1	1	0	0
10	0	0	0	0

$T_2 = Q_1 Q_0 \bar{M} + \bar{Q}_1 \bar{Q}_0 M$

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Step 5



Synchronous BCD Counter

Synchronous :-

Step 1:-

No of bits = 4-bits

FFs = 4

TFFs

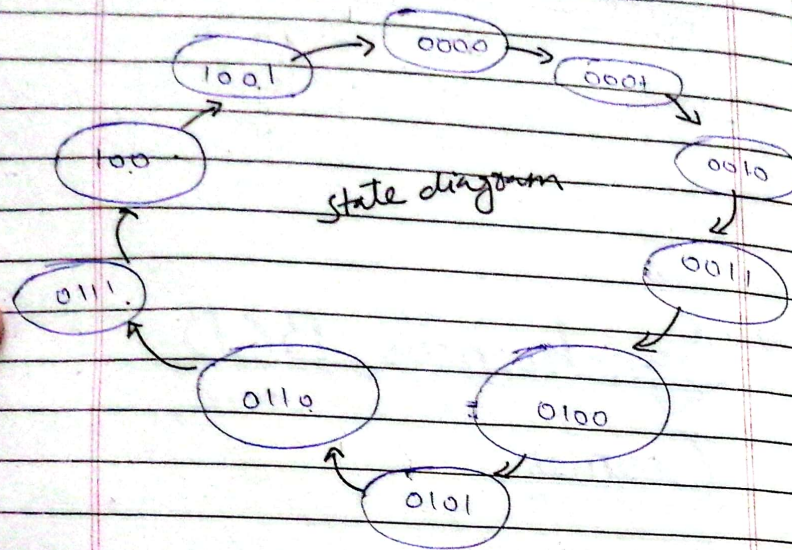
Step 2

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

(27)

Date _____

Step 3:



Step 4 :- PS

NS

$Q_3 Q_2 Q_1 Q_0$	$Q_3 Q_2 Q_1 Q_0$	$T_3 T_2 T_1 T_0$
0 0 0 0	0 0 0 1	0 0 0 1
0 0 0 1	0 0 1 0	0 0 1 1
0 0 1 0	0 0 1 1	0 0 0 1
0 0 1 1	0 1 0 0	0 1 1 1
0 1 0 0	0 1 0 1	0 0 0 1
0 1 0 1	0 1 1 0	0 0 1 1
0 1 1 0	0 1 1 1	0 0 0 1
0 1 1 1	1 0 0 0	1 1 1 1
1 0 0 0	1 0 0 1	0 0 0 1
1 0 0 1	0 0 0 0	1 0 0 1

(28)

Date T3

$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00	00	0	0	0	0
01	00	0	0	1	0
11	00	X	X	X	X
10	00	0	1	X	X

$$T_3 = Q_3 Q_0 + Q_2 Q_1 Q_0$$

$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00	00	0	0	1	0
01	00	0	0	1	0
11	00	X	X	X	X
10	00	0	0	X	X

$$T_2 = Q_1 Q_0$$

$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00	00	0	1	1	0
01	00	0	1	1	0
11	00	X	X	X	X
10	00	0	0	X	X

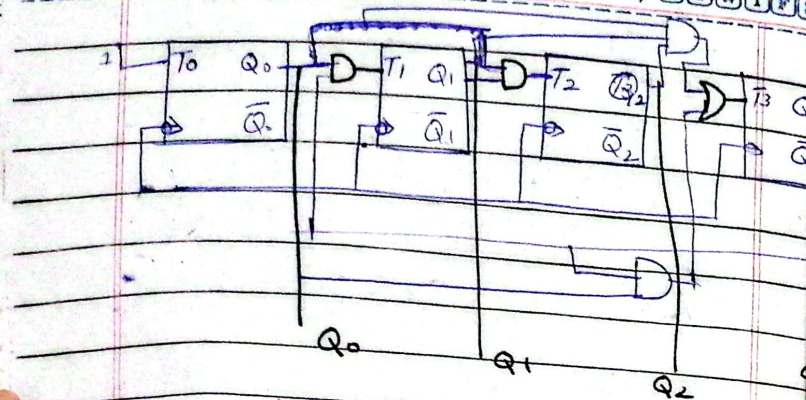
$$T_1 = \bar{Q}_3 Q_0$$

$$T_0 = 1$$

Step 5:

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Date _____

Day MTWTFSS

Synchronous MOD-10 up/down Counter

Step 1:- $N \leq 2^n$

No of bits = 4-bits
4 FFs

$$10 \leq 2^4$$

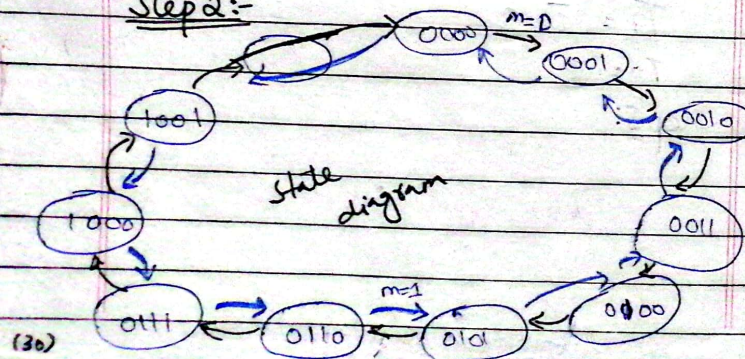
$$\text{max-state} = 2^n - 1 = 16 - 1 = 15 \text{ states}$$

M = 1 - Down counter

M = 0 - Up counter

10 / 6 invalid

Step 2:-



(30)

Step 3:-

Excitation table:-

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 4

	PS				NS				$T_4 T_3 T_2 T_1$			
	Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	T_4	T_3	T_2	T_1
0	0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	1	0	0	1	0	0	0	1	1
0	0	0	1	0	0	0	1	1	0	0	0	1
0	0	0	1	1	0	1	0	0	0	1	1	1
0	0	1	0	0	0	1	0	1	0	0	0	1
0	0	1	0	1	0	1	1	0	0	0	1	1
0	0	1	1	0	0	1	1	1	0	0	0	1
0	0	1	1	1	1	0	0	0	1	1	1	1
0	1	0	0	0	1	0	0	1	0	0	0	1
0	1	0	0	1	0	0	0	0	1	0	0	1

(31)

Date 0000

1001

Day M T W T F S

M	1001	1000	0001
↓	1000	0111	1111
↓	0111	0110	0001
↓	0110	0101	0011
↓	0101	0100	0001
↓	0100	0011	0111
↓	0011	0010	0001
↓	0010	0001	0011
↓	0001	0000	0001

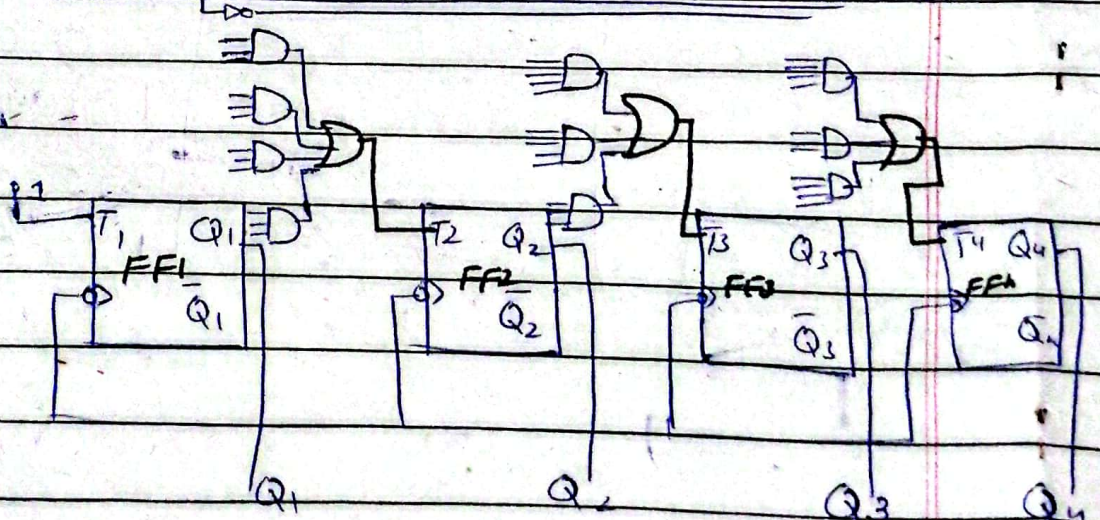
Minimal expression:-

$$T_1 = 1$$

$$T_2 = Q_4 \bar{Q}_1 M + \bar{Q}_4 Q_1 M + Q_2 \bar{Q}_1 \bar{M} + Q_2 \bar{Q}_1 M$$

$$T_3 = Q_3 \bar{Q}_2 \bar{Q}_1 \bar{M} + Q_4 \bar{Q}_1 \bar{M} + Q_2 Q_1 M$$

$$T_4 = \bar{Q}_3 \bar{Q}_2 \bar{Q}_1 \bar{M} + Q_4 Q_1 M + Q_3 Q_2 Q_1 M$$



(32)